

Quan (Ethan) Yuan

quanyuan2028@u.northwestern.edu | (+1)447-446-0255

EDUCATION BACKGROUND

Northwestern University

BS in Computer Engineering

Evanston, IL

Sept 2026 - Jun 2028 (Expected)

University of Illinois Urbana Champaign

BS in Computer Engineering

Champaign, IL

Aug 2024 - May 2026

Overall GPA: 4.0/4.0

Award & Honors: Dean's List, James Scholar

EXPERIENCE

Computer Vision & Graphics (3D Gaussian Splatting), Advisor: Prof Sarita Adve

Champaign, IL

Undergraduate Research Assistant

May 2025 - Aug 2025

- Built a CUDA/PyTorch pipeline to compute per-Gaussian contribution scores (opacity, volume, spatial metrics) for selective rendering
- Implemented GPU kernels to sort Gaussians by contribution and prune low-impact points to minimize rasterization cost
- Integrated contribution-based filtering into the 3DGS renderer, enabling progressive rendering with lower compute overhead while preserving visual quality

Illini Solar Car

Champaign, IL

Member, Electrical Team

Sept 2024 - Aug 2025

- Upgraded the vehicle's brake and control systems by implementing and refining embedded C++ control logic to improve system reliability and responsiveness
- Designed, assembled, and soldered multiple custom PCBs for the control subsystem, including signal routing, power regulation, and interface connections
- Collaborated with cross-functional mechanical and electrical teams to integrate subsystems into the vehicle; contributed to achieving **1st place** in the SOV Class of the Formula Sun Grand Prix 2025

ECE 120: Introduction to Computing & ECE 220: Computer Systems and Programming

Champaign, IL

Undergraduate Grader

Sept 2025 - May 2026

- Evaluated and graded C/C++ programming assignments, digital logic homework, quizzes, and midterm exams for ECE 120 and ECE 220 courses
- Reviewed student code to identify logic errors, inefficiencies, and optimization opportunities while providing constructive feedback on debugging techniques
- Collaborated with instructors and teaching assistants to refine grading standards and maintain consistency across the teaching team

PROJECTS

FPGA Terraria-inspired Game

Feb 2026 - May 2026

- Developed a Terraria-inspired side-scrolling game on an FPGA using SystemVerilog, MicroBlaze C, AXI4-Lite, HDMI output, and USB keyboard input
- Implemented real-time hardware rendering for scrolling backgrounds, tile maps, animated sprites, HUD elements, start/death screens, enemies, item drops, and boss graphics
- Built software game logic for player movement, collision detection, mining, block placement, inventory tracking, health system, enemy AI, knockback, and boss behavior
- Selected for the **ECE 385 Final Project Showcase** as one of approximately **13** projects out of **200+** teams

Embedded System Projects (using STM32)

Sept 2023 - Aug 2024

- Developed microcontroller-based embedded systems in C, including an OLED display game, Mecanum-wheel car, and magnetic levitation system
- Implemented SPI/I2C communication, integrated sensors and actuators, and designed custom PCBs for embedded control systems
- Built control logic for wireless remote control, autonomous obstacle avoidance, magnetic levitation, and real-time display feedback

SKILLS

- **Programming language:** C, C++, Java, Python, Assembly, R, SystemVerilog
- **Software:** KiCad, MCUXpresso IDE, Android Studio, Keil uVision, Fusion 360, Vivado, Vitis
- **Instrumentation:** Oscilloscope, Logic Analyzer, Digital Multimeter, Function Generator